

# ML Model: Rented Bikes Demand in Seoul

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**Abstract**—This project develops a machine learning model to predict hourly bike rental demand in Seoul, using the Seoul Bike Sharing Demand dataset donated to the UC Irvine Machine Learning Repository, which contains 8760 hourly instances chronologically ordered from December first 2017 to the 30 of November 2018. The process of the project will start with an ETL method to treat the raw data in which you can access the notebook on the repository to follow the step by step I made. Then the data is submitted to an exploratory data analysis (EDA) in which the characteristics of demand, behavior of the dependent variables and the limitations imposed by a seasonality based data split is analyzed. The model development begins with a linear regression implemented from scratch using batch gradient descent which serves as a baseline that establishes what a purely linear relationship can achieve in this problem. The EDA reveals cyclical behaviors and threshold effects that prove the linear hypothesis can not represent this problem. Then three framework models are built for comparison: an ordinary least squares regression (OLS) that will verify the manual implementation and provide the statistical significance to each coefficient to declare the efficiency of the model, a decision tree and a random forest. All of the models are trained on the identical transformed training subset and evaluated with the same metrics against the linear model as baseline. The linear models explain roughly about 35% with a theoretical maximum explored of 40% of the variance of the hourly rental bike demand while the random forest reaches 89%, isolating the linear assumption rather than the data split as the limiting factor.

**Index Terms**—Machine Learning, ETL, EDA, Seoul Bike Sharing Demand, Prediction Model

## I. INTRODUCTION

Bike-sharing systems depend on anticipating the demand. Knowing how many bicycles are expected to be requested in given hours determines how units are redistributed across stations, when maintenance can be scheduled without disrupting the service and an expected output of usage that can translate to profit. Demand, however, is not driven by a single factor, it responds to the working calendar, to the wide weather conditions and interact with each other rather than acting independently. This project addresses that prediction problem using the Seoul Bike Sharing Demand data sets which contains hourly records of the bike rentals over one year including other variables as weather conditions. The objective is to predict the number of rented bikes given the remaining measurable features, and to determine which family of models is appropriate for the structure of this particular problem. The work is organized as follows: Section II documents the ETL process and the decisions taken to make the raw data usable, including the division of subsets. Section III presents

the exploratory analysis, which characterizes the demand and assesses whether the chosen split preserves the information the model needs. Section IV justifies the selection of models, and Sections V to VIII report the construction and results of each one. Section IX compares them and concludes.

It is possible to evaluate this problem and make a prediction through pure statistics; however this project looks forward to taking advantage of machine learning and producing an accurate prediction. Although the development and results are limited and confined by the data itself including the window of time, it should be a first step in understanding an introduction to machine learning for me and the reader. If someday an absolute model were created that could predict bike demand with such precision for Seoul or other cities, this and all work on the same data and similar projects would be the first step.

## II. EXTRACT, TRANSFORM, LOAD (ETL)

### A. Data Extraction, Initial Inspection and Understanding

To produce an ETL we need to understand the raw dataset. The dataset comprises 14 features and 8760 instances in particular. We are talking about 365 days from 1 December 2017 to 30 November 2018, divided by hour instances; besides, out of those 14 features, the goal of the project is to predict the feature Rented Bike Count given the other parameters.

TABLE I  
RAW DATASET DESCRIPTION

| Feature               | Description                                           | Type    |
|-----------------------|-------------------------------------------------------|---------|
| Date                  | Format dd/mm/yyyy                                     | Object  |
| Rented Bike Count     | Total bikes rented per hour (Target variable)         | Integer |
| Hour                  | Hour of the day (0–23)                                | Integer |
| Temperature           | Ambient temperature (°C)                              | Float   |
| Humidity              | Relative humidity (%)                                 | Integer |
| Wind speed            | Wind speed (m/s)                                      | Float   |
| Visibility            | Visibility distance in 10m units                      | Integer |
| Dew point temperature | Temperature at which air becomes fully saturated (°C) | Float   |
| Solar Radiation       | Accumulated solar energy ( $MJ/m^2$ )                 | Float   |
| Rainfall              | Rainfall amount (mm)                                  | Float   |
| Snowfall              | Snowfall amount (cm)                                  | Float   |
| Seasons               | Seasonal periods of the year                          | Object  |
| Holiday               | Holiday indicator (Holiday / No Holiday)              | Object  |
| Functioning Day       | Operational status of the rental system               | Object  |

Furthermore, for these variables it is important to understand that: In the feature Visibility, the method of measurement

or instrument is unknown, but every unit is expressed as "10m" referring to a longitudinal distance in which it is visible, probably due to weather factors. The Dew point is the temperature at which air becomes fully saturated with water vapor and cannot hold moisture any longer. It is important when predicting weather forecasts due to its relation with other factors. Solar Radiation is measured in MegaJoules per square meter and represents the total accumulated energy produced by the sun in a lapse of time (Hour in this case). The Date is resourceful to understand results and the chronology of events; however, it won't be fed to the model since it isn't helpful for a prediction, just for a search. Worth mentioning that the dataset had no null values for any feature.

### *B. Understandable Data for the Model*

The first step is the transformation of all Objects and Strings into understandable data. For Holiday, we transformed it into a binary integer value; it either is a holiday or isn't. Seasons were transformed into an integer variable, 1-4, starting with Winter to try and make a numerical distinction. Functioning day was also transformed into a binary integer value.

### *C. Visibility Uncertainty Translation*

For the visibility uncertainty on how it was measured, it was translated into a ratio of how much was visible for users when riding a bike, where 0 is no visibility and 1 is the maximum visibility; this was for better digestion for our model. This could be done because, after data inspection, it had a maximum value of 2000 and referred to a distance value, so it could be scaled.

### *D. Non Functioning Day Imputation*

By nature, although having no null values, non-functioning days had a rented bike count of 0. After inspection, there were 295 non-functioning hours that represent approximately 3.36% of the dataset, and only these 295 instances presented an absolute 0 rented bike count. Due to the assumption that in the dataset there is not another possible combination that gives a rented bike count of 0 and represents a small amount of information, it was decided to perform an imputation of the rented bike count for these instances. The method for imputation was interpolation of the data. The interpolation was calculated as the rounded mean value for the same hour from the closest functioning instance before and after. The original zeros could not be fed to the model because it would learn an obvious trivial rule that a non-functioning day will produce 0 rentals, and as is our intention to exclude it of all subsets, that pattern will instead be attributed to whatever conditions happened to coincide with those instances. They were equally useless for evaluation purposes since a prediction can not meaningfully be compared against a value that reflects the absence of service rather than the absence of demand. The imputed values replace them by creating an artificial value that the model can learn latent demand rather than observed supply; the affected instances remain identifiable through the map subset for later diagnosis.

### *E. Date Feature Compensation*

Although the date itself will not be fed to the model, a binary value can be extracted from it to help the model capture weekly routines. From the date, using pandas `pd.to_datetime`, we can get an int binary value for the weekend feature.

### *F. ETL Dataset*

Up to this point, we do not possess null instances across all the data. The data has been transformed into digestible, understandable units and values. All calculations for the new data were successful and do not contain errors. The data was treated to the optimal type. Finally, all instances are still in chronological order. We end with this post-ETL dataset.

### *G. Division of Classes*

We can now produce our division of classes for the model. The division percentage is set to be 90% of the data destined to train and the remaining 10% to test and validate the model. To split the data, we have to consider the factor known as seasonality or trend. In finance and economics is best noticed by the annual seasons in which data patterns emerge and behave differently between seasons. Therefore, it was decided that the division of data will be divided equally from all seasons. 36 days, representing approximately 10% of the total data, will be taken from the last days of each season, meaning that for each season the last 9 days, or 216 instances, will be destined for the holdout. The Y train and test subset will contain only Rented Bike Count; the X subset will have the rest, excluding the Date and Functioning Day. Recycling these features of data we create also a subset defined as Map that will contain these reused excluded features, as well as all the instances in order with the corresponding season, the subset it belongs to (train, validation or test), and a local ID for the subset. This is with the purpose of having a map that will act as a tool when processing results, to graph all data in chronological order and explain possible anomalies due to the imputation or division of continuous data. The justification for creating the map is that it provides the information to order and filter data. After the division, we are left with the next subsets of data. Finally, the remaining 10% will be split into 120 instances for testing and 96 for validation; the test set gets the last 5 days of the season and the validation the 4 days before this.

## III. EXPLORATORY DATA ANALYSIS (EDA)

The EDA of interest only falls under the training features, and on this occasion we will include our dependent variable. At this point, we can produce copies of our subset to model at our will without losing the information for the model.

### *A. Distribution with Histogram*

For the analysis of histograms, we discriminate the data for Hours, Seasons, and Weekend, as that categorical data frequency won't add information. We then can produce of all histograms for all continuous data, including the histogram for Holiday. We are left with the following graphs.

TABLE II  
POST ETL DATASET

| Feature               | Type        | Feature Class |
|-----------------------|-------------|---------------|
| Date                  | Datetime    | Map           |
| Rented Bike Count     | Integer     | Y             |
| Hour                  | Integer     | X             |
| Temperature           | Float       | X             |
| Humidity              | Integer     | X             |
| Wind speed            | Float       | X             |
| Visibility            | Float,Ratio | X             |
| Dew point temperature | Float       | X             |
| Solar Radiation       | Float       | X             |
| Rainfall              | Float       | X             |
| Snowfall              | Float       | X             |
| Seasons               | Int         | X             |
| Holiday               | Int,Binary  | X             |
| Functioning Day       | Int,Binary  | MAP           |
| Weekend               | Int,Binary  | X             |

TABLE III  
REPRESENTATION OF DIVISION OF DATA

| Class/Subset | Description                                                      |
|--------------|------------------------------------------------------------------|
| X test       | Last X features from the last 120 instances from each season     |
| Y test       | Last Y features from the last 120 instances from each season     |
| X val        | First 96 X features from the last 216 instances from each season |
| Y val        | First 96 Y features from the last 216 instances from each season |
| X train      | Left X features from the split from each season                  |
| Y train      | Left Y features from the split from each season                  |
| Map          | All map features sorted by an index, keeping chronological order |

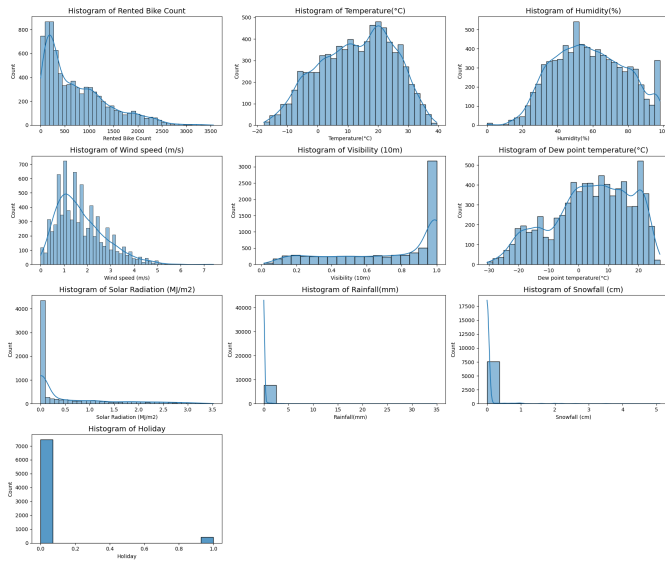


Fig. 1. Histograms of continuous variables and holiday distribution.

**1.1** The distribution for the Rented Bike Count is shown as a positively skewed distribution with a long tail. Most of the data falls below 500; we can assume that most instances or hours present fewer than 500 rentals.

**1.2** The distribution of Temperature seems a bit symmetrical, with a higher frequency at approximately 20 degrees. Most instances during the year of data collected were about 20 degrees Celsius in Seoul.

**1.3** The distribution of humidity, although it almost looks symmetrical it has a high bias towards high values, with only a few occurrences around 0% and a high peak at 100%. With peaks at 50 and 100, we can assume that most instances presented at least a slightly higher humidity percentage in the data collected in Seoul.

**1.4** The wind speed distribution presents a positively skewed distribution with a long tail distribution with most values falling around 1m/s. The majority of instances presented less than 3m/s wind speed readings in Seoul.

**1.5** The distribution of visibility is highly negatively skewed, with the most frequent value being 1 also understood that the majority of instances present 100% or complete of visibility.

**1.6** The distribution of Dew temperature seems to present a non-symmetrical distribution, with no high bias for the left side. It presents two pinnacles: first at around -20 degrees, then falls into a valley around -10 and rises to a higher pinnacle and larger group of frequency around 10 degrees, with a noticeable single peak at 20 degrees. We can assume that it presents a bimodal characteristic and that two groups of data could form around -20 degrees and 10 degrees of dew point; the dew point is ultimately noticed by people as it has a highly physical relation with weather. Therefore, there could be the possibility that it could provide a high numerical relationship when comparing the output with 2 possible classes. It could also be useful that the 20 degree peak has a strong numerical relationship with Temperature.

**1.7** The histogram of solar radiation presents a positively skewed frequency, with most of the data falling near 0 MJ/m<sup>2</sup>; most instances in Seoul present around 0 units of solar radiation, also interpreted as most of the time there are no high readings of energy. This could be a highly noticeable remark to drop the information, as most information containing 0 as a value could not contribute to the model. However, the 0 reading could be explained by the recorded instances happening at night time or being a characteristic of Seoul. We can also expect the possibility of the model creating a relationship between this and other features, meaning that this data could identify possible groups in this highly skewed data and therefore is kept and will be fed to the model.

**1.8** The histogram on rainfall shows most of the values fall near 0, below approximately 2.5. The distribution could be considered to not be a feature to study or even transformed as most of data falls on a small range however we have outliers and a maximum record of 35.0 millimeters in rain, these outliers of high rain instances could provide still a high chance influence in predicting, meaning with small data it can

be almost absolute assumption that high rainfall produces very low rental behavior therefore is kept for the model.

**1.9** The histogram on the snowfall follows a similar behavior, with most of the data falling below 0.5 cm and, due to the same logic, it could be argued that this information can be dropped or transformed; however, the maximum value occurring at 5.1 cm also gives the possibility of a high indicator of rental behaviour. These outliers could play an important role in the prediction model; therefore, it's kept.

**1.10** The histogram on holidays was produced to understand a visual representation of the ratio between non-holiday or workless days. These days are important as it highly impacts the routine of the citizens of Seoul. Important to mention that this data can be fed without transformation; this is a binary value, so there is no use in exploring deeper into the distribution and mean; therefore kept as is for the model. I do propose the consideration that it could be transformed or grouped with more information, or even dropped, to explain, in a deeper manner, the behaviour of holidays. Meaning that holidays could impact in a highly unpredictable way as is, since holidays could produce high rental behaviour or even a lower one due to the possibility on lower citizens being in Seoul during these instances.

## B. Distribution with Boxplot

We now proceed to create box-and-whisker graphs of box and whiskers for all continuous variables; graphing for categorical and discrete variables is not worth doing. We are left with the next graphs. It will be useful to visualize possible outliers better that could not be well noticed on the histogram graphs.

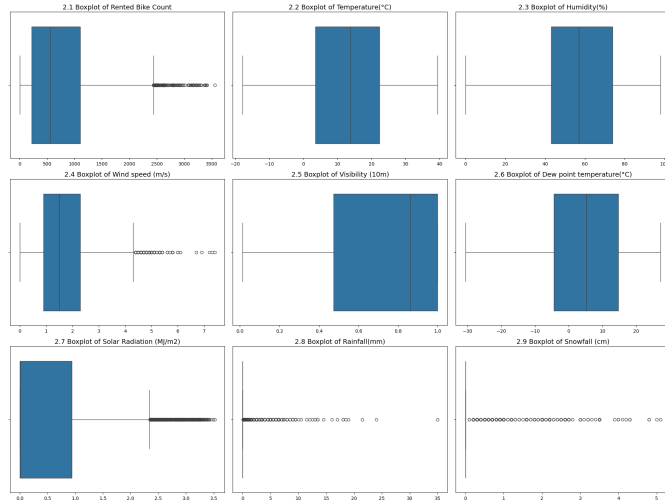


Fig. 2. Boxplot of continuous variables and holiday distribution.

**2.1** To describe it, the box itself is located on the left side, and the median is around 500 counts on the left side of the box. Regarding distribution, it has wide whiskers or spread, with high outliers. The worth-mentioning information to spot on the Boxplot for the Rented Bike Count is the high number of outliers. This could be an early remark that many

real possible specific combinations of X features can provide high rental behaviour.

**2.2 , 2.3 , 2.5 , 2.6** After inspecting the boxplots produced for Temperature, Humidity percentage, Visibility and Dew point temperature, we can state that there are no noticeable outliers. Describing the graphs, it shows a highly packed box distribution in the data. The thing to consider is that the median for Temperature falls around 15 degrees and is located on the half of the box, making it a symmetrical distribution. For Humidity, the median falls around 55%, and it is more on the left side of the box. For Visibility, the length of the box is bigger but still skewed back, with the median located at around 0.85 on the right side of the box. Finally, for Dew point, the box is heavily packed and the median cuts symmetrically through the box, falling around 8 degrees. Even with large whiskers, the lack of outliers and a highly packed box is a sign of a more balanced or spread-out distribution. It indicates that the data is well scattered and most of the data has fallen within a statistical scope. It is important to take this factor in consideration when analysing further data also open to the possibility of falling by nature into a normal distribution, at this point the histograms 1.2, 1.3, 1.5, 1.6 did not present a strong characteristic of normality however it could be an early sign that this is due for the limitations on the data sample and that having a larger sample size of more than 1 year it could be appreciated a normal distribution then having solid new information on how to treat and transform this variables.

**2.4 , 2.7 , 2.8 , 2.9** To describe Wind Speed, the box is located on the left side with a median around 1.5; the box almost looks symmetrical, but the median falls more to the left side. The Solar radiation Box has the median close to the left end of the box, indicating a high concentration of data around 0.0. It is worth mentioning that, in the case of Rainfall and Snowfall, the Box and whiskers are present in this graph; this is because it is a highly concentrated and skewed distribution, so both boxes and whiskers are small and located on the left side of their corresponding graph. Important to keep in mind when analysing further data. When checking the whiskers for Wind Speed. Solar Radiation, Rainfall and Snowfall present a high number of outliers. As mentioned before, all of these points do not mean they are negative characteristics for the model, and they will not spoil it; they are real possible events that could happen in Seoul. The purpose is not to eliminate or impute new values to these outliers, but to understand how they could affect the results of the model and how to manage and interpret the data, as the mean and variance for these values will have another perspective.

**2.10** Shows that each season contains many outliers, which theoretically is not ideal, but it can be broken down. For the winter these outliers are acceptable and understandable: the instances are hourly, and the difference in factors between early hours and midday can be drastic, so it does not follow that they will be extreme to predict. Besides the box plot itself is very compact and the whiskers are not that large, which sustains that the dispersion could be still acceptable. For Autumn it appears to contain the fewest outliers out of them

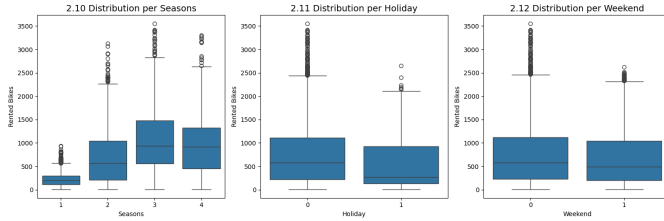


Fig. 3. Boxplot of categorical features.

all and this is also acceptable; the whiskers although large screaming reasonable. The box itself is large and the mean splits its center, which again indicates a healthy dispersion. For spring and summer, the high number of outliers and the long whiskers do raise the suspicion of whether rental demand spikes or is widely distributed. The box also falls far below the highest values, which could be a sign that this wide dispersion will make the linear regression model difficult. **2.11** Give us a great result even if the non-holidays seem to be more dispersed due to the high number of outliers; the holiday category does seem to be more docile to our understanding and could be a sign that the model could recognize a pattern better on holidays. We shall also recognize that the holiday itself, even if it seems chaotic, compresses each instance of hours during a day and many more data; therefore, it's not a full sign it will be that chaotic for the model.

**2.12** Take into consideration that the variable weekend was created; this boxplot gives the first insight if the variable itself works. It does tell the same story as the distribution per holiday; this could be evidence that the variable categorizes important behaviours; however, we should dig deeper to prove this.

### C. Behaviour through line plots

We will average every hour's demand across all of the train subset for every hour in the day. The purpose of this graph is to know if this feature can be useful. Reading the graph, we can notice some similarities across early hours and the middle of the day; however, the spike readings on weekend days with high demand at 8 am and 18 pm hours create the importance of maintaining this feature. Therefore, this feature won't be dropped.

We can apply this same code logic and produce line plots for every feature, telling with the best perception how each behaviour affects rental bikes across an average day. This is an important visualization that lets us better comprehend the individual effect each one has. The nature of the code needs a metric and requires declaring classes and bins to distinguish continuous data. Therefore I adjusted an approximation so each value of each feature can belong to a corresponding class, imitating the fuzzy class concept, although the implementation criteria differed for each. Rainfall, solar radiation and snowfall were turned into binary outputs to produce an easier visual graph and a more general depiction of the behaviour: if the reading is zero the event is not occurring, and otherwise it is, which lets us illustrate bike demand for when each of these

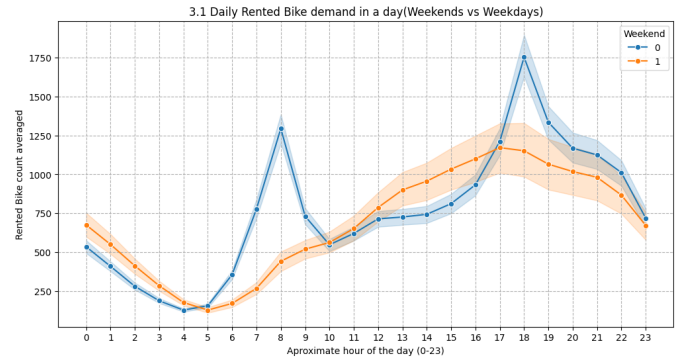


Fig. 4. Average demand for bikes during a day regarding the weekend feature.

conditions is present or absent at a given hour of an average day. For categorical data such as Holiday and Seasons, the existing values already group into classes for the line plot. The remaining continuous data was matched into three classes describing a minimum reading, a mild reading and a high value, with bins established from the maximum and minimum of each feature.

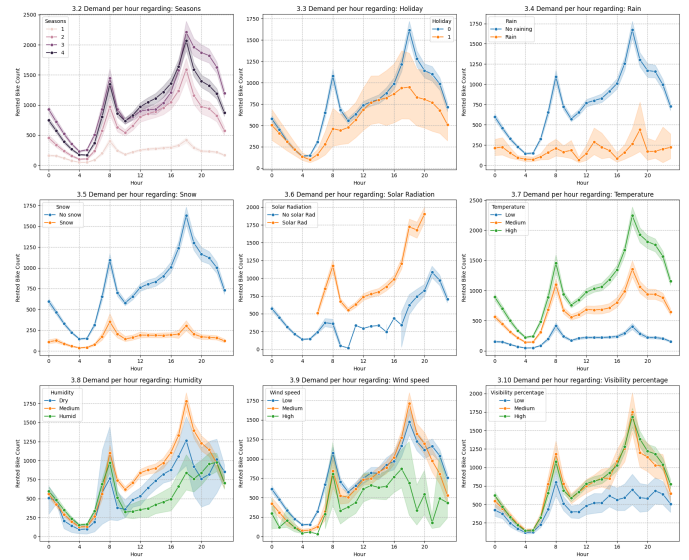


Fig. 5. Average demand for bikes during a day regarding other features.

**General findings** In all days, no matter what the feature is or the values of the features, it always appears the same two spikes about 8-9 and 17-18, except for the holiday graph. This surely marks the behaviour of the labour schedule that heavily impacts the prediction of the bikes. To not leave things pending, I double-checked, and by different means I corroborated that the usual work shift in Seoul is a 9 to 18 hour shift. With this finding, I do raise the consideration that it would be highly impactful to introduce a way it has more importance to these instances, choose a model that can perceive and distinguish this hourly peak and not generalize across instances or create another split arrangement.

**3.2 , 3.3** The behaviour in the seasons and seasonality

effect already gave us information; however, this graph depicts a different story. Zooming in on the average for every hour in the day tells us that, with differences, winter has less demand overall. On an average day, spring, summer, and autumn look similar, with a few hundred bikes' difference. The holiday does confirm there is an effect for rentals: the non-working days have less demand and soften the hourly workshift peaks.

**3.4 , 3.5 , 3.6** For rain, snow, and solar radiation, it gives us an approach to the expected behaviour. On average, rainy instances pick up fewer rented bike counts compared to non-rainy instances, and this is the same behaviour for snowing vs not snowing instances. Now, to explain the radiation, it now shifts the no solar radiation instances have, on average, fewer rental bike counts, and radiation days do have more. What is key to understand is that this is binary; it doesn't tell us how much radiation is fluctuating specifically, so it is plausible that on non-solar radiation instances there is less count; it's because it can be correlated with heavy, windy, cold, or rainy bad weather. What we do can learn is that, from solar radiation instances, it could possibly match better if we understand better the average daily cycle of the solar radiation; comparing the A.1 graph, we can visualize that after the greatest peak of solar radiation happening at approximately 13 hours, it starts to grow the demand on bikes as it falls down.

**3.7 , 3.8 , 3.9 , 3.10** From temperature, humidity, wind speed, and visibility percentage, we observe three classes behaving similarly, with some important phenomena interesting to explain. I realized that, similar to solar radiation, it is also expected that Temperature can behave in a similar cyclic behaviour, so I created an average daily temperature cycle graph to realize that it could be explained and does fit the story that the high temperatures around 3 pm and the downfall to a low, not extreme temperature at noon fit the growing tendency of the rental demand in the same time lapse. For humidity, the most notable characteristic is that, in the early hours, the difference in demand between classes is similar, but after 9 am, medium humidity has the majority of demand, followed by the driest instances and then the expected humid instances. Then, for wind speed, all instances in bike demand are close to each other, but it seems that after 4 pm, high wind speed gives lower levels of bike count. Then, for visibility, it seems that low visibility does give, in general, lower demand behaviour after 8 am, even decreasing at the end of the work shift to a difference of approximately a drop of 1000 counts.

#### D. Time series analysis

The general analysis on the time series for the rental bike was at first just a proposal to prove this dataset did have seasonality; however, it was important to re-graph it after the split to better understand what we were taking away from the train subset, as the important thing is to keep the most representative data. The comparison was made through line plot and scatter plots.

**4.1** We will now analyse the average daily bike demand over the year of data for only the Y train subset, leaving the gaps for test and val. This is made simpler through the map

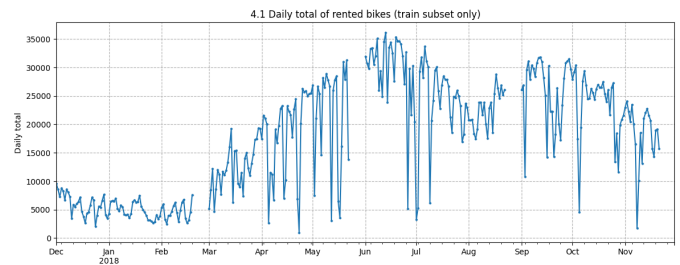


Fig. 6. Daily rental count across a year for train subset.

subset to know what periods of time in order to graph. It is now possible to appreciate a difference between seasons in the daily demand. Winter (December through March) has the lowest daily average, and Summer (June through September) has the highest average daily peaks of the year. Describing the whole year: December starts slow, and March starts growing with considerable large downfalls, but the growth gets to the max peak in June; then it starts to descend to August around 1000 rentals and then rise through September and October before falling again, arriving at November.

Then we arrived at the comparison with the scatter plot and all subsets, and I noticed some outliers from the non-training subsets, which created this next section.

#### E. The elephant in the room. Outliers in the split problem.

The question I proposed was Does the split of the subsets take away important information to train the model?

To understand this, we can take the 4.2 graph and visualize clearly the contents of each subset. In comparison, in the 4.1 graph, we disregard the test and validation subsets so we can better appreciate that there are some big outliers that break a pattern, with huge sudden changes increasing or decreasing considerably from one day to the next. Meaning that any other split considered could also have the same problem of low accuracy due to the model lacking these outliers to train and having problems predicting or training with these outliers and changing the parameters' final adjustments. We then produce

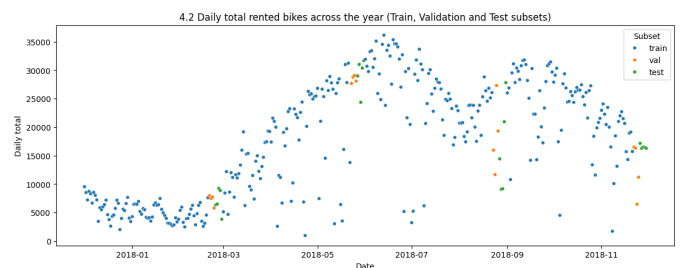


Fig. 7. Daily rental count across a year for all subsets.

the 4.2 graph comparing all three subsets, we can notice most of the outliers that the train doesn't get will happen around september this can be acceptable due the majority of data that wont predict do adjust better to the noticeable pattern. We have to take into consideration that this is only for the resulting data,

or the data we want to predict, so we only used it as a first indication of possible bias by omission based purely on the results. In order to fully conclude whether there can be this danger, we have to study the X features further and notice that there is actual information left behind. So far, we can notice that after September there will be a high amount of anomalies; we then ask the question Is there any of the data that justify those changes? If so, can we prove it can be picked up in the same pattern from just the training information?. To answer this, we will consider that similar statistical outcomes across the 3 subsets will produce the same results regardless of the split decision. Differences in the statistical information of the subsets could correlate with the anomalies observed; therefore, the possibility that the model can pick up on this.

#### F. Outliers Detection.

We will find outliers in our X variables from all subsets by analyzing tables IV and V, the mean, standard deviation, maximums, and minimums. First, we do a general inspection with these operations for all variables. We are looking at a possibly important outlier if the difference between subset stats differs considerably. Having mentioned that, we find our first suspect for further study

Outlier criteria. The outlier criteria I proposed is that only 1 discrepancy does not raise suspicion; the discrepancies in the train subset are less important than the validation and test discrepancies. It can be a subject of study even if it contains only 1 discrepancy if it has a substantial difference between subsets.

- Temperature: the general mean is fairly the same across the subsets; the std is equally high for all of them, but the maximum and minimum trigger the warning. The train contains the coldest instance, with the difference having a minimum of -17.8 degrees Celsius, and the test and validation only reach a minimum of -3.1 and -4.3 degrees Celsius, respectively. The train also contains the hottest temperature recorded, with a difference of 39.4 degrees Celsius, while test and val rise to 30.3 and 32, respectively. In theory it is ideal that the train subsets gets the greatest outliers so it could possibly learn from this and then test and validation can be under these confinements but reality does not always fit into therefore the discrepancy makes it a subject of study.
- Humidity gives fairly close outcomes for mean, std, and maximum, with only one minimum discrepancy in the validation subset, never getting a completely dry instance (14% being the driest instance) in comparison to the other that do get a 0. This I consider to be a small discrepancy; therefore, it is not subject to further study.
- Wind speed produces similar outcomes across all stats in exception for train having the windiest instance being 7.4 meters per second in comparison to 4.3 and 5.3, its considerable difference taking into account that the values only ranges from 0 to 7.4 however the data that tell us something to study is the Validation as the mean

TABLE IV  
DESCRIPTIVE STATISTICS FOR ALL SUBSETS.

|                           | Subset | test   | train  | val    |
|---------------------------|--------|--------|--------|--------|
| Temperature(°C)           | mean   | 13.42  | 12.88  | 12.26  |
|                           | std    | 9.91   | 12.10  | 11.03  |
|                           | min    | -3.10  | -17.80 | -4.30  |
|                           | max    | 30.30  | 39.40  | 32.00  |
| Humidity(%)               | mean   | 60.12  | 58.17  | 57.07  |
|                           | std    | 23.74  | 20.00  | 22.95  |
|                           | min    | 0.00   | 0.00   | 14.00  |
|                           | max    | 98.00  | 98.00  | 97.00  |
| Wind speed (m/s)          | mean   | 1.60   | 1.72   | 1.96   |
|                           | std    | 0.91   | 1.04   | 1.04   |
|                           | min    | 0.00   | 0.00   | 0.00   |
|                           | max    | 4.30   | 7.40   | 5.30   |
| Visibility (10m)          | mean   | 0.61   | 0.72   | 0.74   |
|                           | std    | 0.29   | 0.30   | 0.32   |
|                           | min    | 0.04   | 0.01   | 0.02   |
|                           | max    | 1.00   | 1.00   | 1.00   |
| Dew point temperature(°C) | mean   | 4.98   | 4.08   | 2.82   |
|                           | std    | 12.79  | 13.08  | 12.92  |
|                           | min    | -19.00 | -30.60 | -22.00 |
|                           | max    | 24.50  | 27.20  | 23.60  |
| Solar Radiation (MJ/m2)   | mean   | 0.50   | 0.57   | 0.61   |
|                           | std    | 0.82   | 0.87   | 0.91   |
|                           | min    | 0.00   | 0.00   | 0.00   |
|                           | max    | 3.34   | 3.52   | 3.52   |
| Rainfall(mm)              | mean   | 0.40   | 0.14   | 0.10   |
|                           | std    | 2.23   | 1.04   | 0.60   |
|                           | min    | 0.00   | 0.00   | 0.00   |
|                           | max    | 29.50  | 35.00  | 9.10   |
| Snowfall (cm)             | mean   | 0.01   | 0.06   | 0.43   |
|                           | std    | 0.04   | 0.35   | 1.28   |
|                           | min    | 0.00   | 0.00   | 0.00   |
|                           | max    | 0.30   | 5.10   | 8.80   |

its greater than the train by itself, this could indicate more windy instances, we will double check.

- Visibility produces similar results across all stats, not a subject of study.
- Dew point temperature almost provides similar results with an exception in the minimum temperature instances that occurred during train being -30.6 in comparison to test and validation being -19 and -22. Also the mean for validation falls under with 2.8 celsius degrees not that far from test and train being 4.98 and 4.08. It is not considered for further study.
- Solar radiation outcomes give very similar results in all specs, not a subject to study.
- Rainfall at first look seemed reasonable, with a considerable discrepancy. The maximum rainfall instance happens in train with a 35 mm rainfall anomaly, and in addition, the reason it is going to be further studied is that the test mean gets 0.40mm; as little as it seems, it triples the mean on train and quadruples val mean.
- In snowfall its important to pay attention in if is there a risk on train not getting snow instances. The mean for train is 0.06 cm which is low in comparison to validation that has an average 0.43 cm and it turns meaningful comparing their respective std being 0.35 for train and a larger 1.28 std for validation. We can also look at maximum values that contains a considerable difference

being 0.3, 5.1 and 8.8 (test, train and validation) cm of snow it will be a subject of study.

We found to be Temperature(°C), Rainfall(mm), Snowfall (cm), Wind speed (m/s) the most dangerous data split since train and validation contains key information of the feature. Therefore we will scope further into by grouping them and comparing by seasons. If the values are not a symptom of seasonality then we can just classify them as expected outliers normal in weather forecast and difficult to predict and won't be looked further into. If it is a cause by seasonality and it got hit right into the split then it should be concluded that there can be another rearrange of data split for future considerations.

TABLE V: Critical statistical values for all subsets

| Variable                          | Seasons | Subset | mean  | std  | min    | max   |
|-----------------------------------|---------|--------|-------|------|--------|-------|
| Temperature( $^{\circ}\text{C}$ ) | 1       | test   | 3.25  | 3.17 | -3.10  | 10.30 |
|                                   |         | train  | -3.09 | 5.38 | -17.80 | 8.70  |
|                                   |         | val    | 1.28  | 3.44 | -4.30  | 8.30  |
|                                   | 2       | test   | 21.03 | 3.44 | 14.30  | 28.70 |
|                                   |         | train  | 12.27 | 6.37 | -6.60  | 29.10 |
|                                   |         | val    | 19.16 | 4.58 | 11.30  | 29.40 |
|                                   | 3       | test   | 24.45 | 2.05 | 20.30  | 30.30 |
|                                   |         | train  | 26.75 | 4.80 | 16.30  | 39.40 |
|                                   |         | val    | 25.69 | 2.99 | 20.40  | 32.00 |
|                                   | 4       | test   | 4.95  | 3.42 | -1.50  | 13.20 |
|                                   |         | train  | 15.23 | 6.46 | 0.30   | 30.50 |
|                                   |         | val    | 2.90  | 2.83 | -3.00  | 9.50  |
| Rainfall(mm)                      | 1       | test   | 0.19  | 1.25 | 0.00   | 9.50  |
|                                   |         | train  | 0.02  | 0.31 | 0.00   | 9.50  |
|                                   |         | val    | 0.04  | 0.38 | 0.00   | 3.70  |
|                                   | 2       | test   | 0.01  | 0.09 | 0.00   | 1.00  |
|                                   |         | train  | 0.20  | 1.26 | 0.00   | 35.00 |
|                                   |         | val    | 0.10  | 0.56 | 0.00   | 3.50  |
|                                   | 3       | test   | 1.40  | 4.13 | 0.00   | 29.50 |
|                                   |         | train  | 0.19  | 1.30 | 0.00   | 24.00 |
|                                   |         | val    | 0.13  | 0.34 | 0.00   | 1.50  |
|                                   | 4       | test   | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | train  | 0.13  | 0.98 | 0.00   | 18.00 |
|                                   |         | val    | 0.11  | 0.94 | 0.00   | 9.10  |
| Snowfall (cm)                     | 1       | test   | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | train  | 0.25  | 0.68 | 0.00   | 5.10  |
|                                   |         | val    | 0.47  | 1.23 | 0.00   | 4.10  |
|                                   | 2       | test   | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | train  | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | val    | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   | 3       | test   | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | train  | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | val    | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   | 4       | test   | 0.02  | 0.08 | 0.00   | 0.30  |
|                                   |         | train  | 0.00  | 0.00 | 0.00   | 0.00  |
|                                   |         | val    | 1.25  | 2.01 | 0.00   | 8.80  |
| Wind speed (m/s)                  | 1       | test   | 1.70  | 0.87 | 0.10   | 3.80  |
|                                   |         | train  | 1.92  | 1.17 | 0.00   | 6.70  |
|                                   |         | val    | 2.19  | 1.20 | 0.00   | 5.00  |
|                                   | 2       | test   | 1.81  | 0.89 | 0.30   | 4.20  |
|                                   |         | train  | 1.87  | 1.08 | 0.10   | 7.40  |
|                                   |         | val    | 2.03  | 1.15 | 0.40   | 5.30  |
|                                   | 3       | test   | 1.54  | 1.08 | 0.00   | 4.30  |
|                                   |         | train  | 1.60  | 0.90 | 0.00   | 6.90  |
|                                   |         | val    | 1.99  | 0.87 | 0.10   | 4.50  |
|                                   | 4       | test   | 1.34  | 0.71 | 0.20   | 3.30  |
|                                   |         | train  | 1.50  | 0.94 | 0.00   | 5.70  |
|                                   |         | val    | 1.63  | 0.81 | 0.40   | 3.40  |

TABLE VI  
OUTLIERS FOR RAINFALL AND SNOWFALL ACROSS ALL SUBSETS AND SEASONS.

| Seasons | Subset | Rainfall(mm) |            | Snowfall (cm) |            |
|---------|--------|--------------|------------|---------------|------------|
|         |        | max          | % of hours | max           | % of hours |
| 1       | test   | 9.50         | 2.50       | 0.00          | 0.00       |
|         | train  | 9.50         | 1.90       | 5.10          | 19.40      |
|         | val    | 3.70         | 2.10       | 4.10          | 14.60      |
| 2       | test   | 1.00         | 0.80       | 0.00          | 0.00       |
|         | train  | 35.00        | 9.30       | 0.00          | 0.00       |
|         | val    | 3.50         | 4.20       | 0.00          | 0.00       |
| 3       | test   | 29.50        | 34.20      | 0.00          | 0.00       |
|         | train  | 24.00        | 6.80       | 0.00          | 0.00       |
|         | val    | 1.50         | 21.90      | 0.00          | 0.00       |
| 4       | test   | 0.00         | 0.00       | 0.30          | 8.30       |
|         | train  | 18.00        | 4.90       | 0.00          | 0.00       |
|         | val    | 9.10         | 2.10       | 8.80          | 42.70      |

Scoping even further, we can then get the extreme values for rainfall and snowfall in all sub-sets in all seasons. Reading our VI table, we have spotted some important outliers. Reading the table for winter, the most important thing is that we can see the maximum value falls under train, which is good; however, we can spot that the frequency of each instance in train is lower. For spring, it does gets the rainiest instances, with a frequency of 9.3 that surpasses test by a lot. Test and validation this season could throw some models, as it could be expecting the rainiest instances around spring and then validating and testing under- or overestimating. For summer, it worries the other way around: validation and test get more than 20% of the instances of rain. Lastly, for autumn, the train does get the maximum value and highest frequency, making it the most representative subset. For snowfall, we only check winter and autumn, as it only snows during these seasons; for winter, it is reasonable as the train gets the maximum value and the highest frequency; however, in autumn it provides an interesting signal. In autumn, the train does not get any snowfall; test and validation do, and even more extreme, validation has a maximum of 8.8 cm, the highest recording in all data set, with more than 42% of instances snowing. The general conclusion is that even with a seasonality split, the train does not always get the most representative content, and the question then is: is it really enough provided to the model to pick up on this data?.

Finally, looking at our Table VII we have created the same analysis system, reading by season by subset the statistics. The biggest difference and most worrisome is the train subset for Spring, the mean of rental bikes falls behind by a lot compared with test and val for the same season; the std also falls behind. Then the greatest notice is that in Summer the training subset does get the higher mean; it could mean that it represents the higher outliers.

#### G. EDA Closure.

We will now try to give a conclusion based on all the results above. First, we have spotted our most problematic features that, due to the split, may present trouble; we can visualize our individual data collected better and try to pinpoint our findings

TABLE VII  
RENTED BIKE COUNT DATA FOR ALL SEASONS AND SUBSETS.

| Seasons | Subset | Statistic measurements |        |     |      |
|---------|--------|------------------------|--------|-----|------|
|         |        | mean                   | std    | min | max  |
| 1       | test   | 289.88                 | 209.44 | 7   | 929  |
|         | train  | 217.81                 | 142.35 | 3   | 937  |
|         | val    | 301.61                 | 179.44 | 32  | 784  |
| 2       | test   | 1220.16                | 740.28 | 134 | 3251 |
|         | train  | 702.50                 | 587.18 | 2   | 3130 |
|         | val    | 1183.24                | 703.37 | 25  | 3123 |
| 3       | test   | 679.06                 | 646.26 | 10  | 2836 |
|         | train  | 1068.03                | 691.68 | 9   | 3556 |
|         | val    | 773.21                 | 516.11 | 30  | 2080 |
| 4       | test   | 689.07                 | 373.12 | 95  | 1708 |
|         | train  | 961.09                 | 627.05 | 2   | 3298 |
|         | val    | 526.03                 | 348.47 | 24  | 1671 |

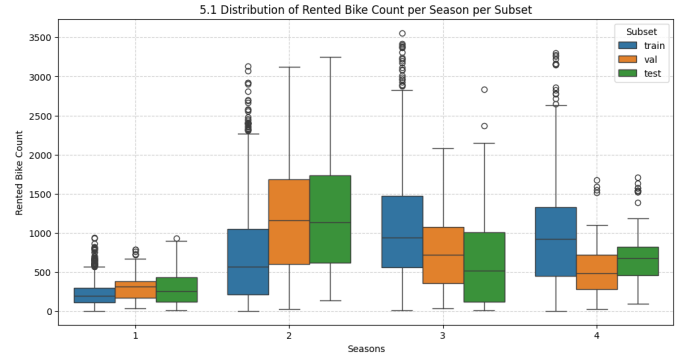


Fig. 8. Boxplot for rental bikes per season per subset.

with graphics 5. 5.1 Repeating the theoretically proposed sweet point is to have the biggest number of outliers in the train section so it demonstrates that it is representative; what we essentially found for the rented bike count was that it was almost true. The problems can be looked in the spring when we notice that even if the train gets the most outliers val and test long whiskers reach these train outliers and for Autumn even if the test and validation falls under the train boxplot this is also not so good we would prefer similar boxes and whiskers lengths as in winters with the fewest outliers possible as in summer.

5.2 , 5.3 , 5.4 , 5.5 For temperature, the overall look is ideal; the data didn't spike before but had to make sure visually as the snow outliers gave me the impression that it could also be affected by temperature, if it wasn't spotted on our tables and visually we conclude the train section does get the representative data and there is not an issue. The rented bike count confirms our reading by season. For wind speed, we find a visually better outcome than expected as the box plots its self and whiskers looks similar and train does get the most outliers, finally the snow and rainfall instances do raise some heavy concern as the bar graph represents the frequency of instances the subset has and for validation it contains a very high frequency of snow, for rainfall it is most acceptable but we also had pinpointed the standard deviation of test rainfall

data was the largest which potentially could indicate a more distribution difficult to predict.

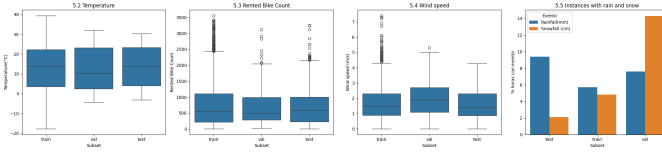


Fig. 9. Boxplot for critical outliers.

We can finally produce a correlation analysis that is represented through this 6.1 heatmap. We are looking for the highest correlation features to try and determine if any of these variables can be dropped due to a high correlation between each. Right off the bat, we can spot the dew temperature and temperature itself with a 0.92 correlation; they move similarly across the whole dataset, and at first it was a logical decision to drop them as they almost target and track similar things. However, I have decided not to drop this entire feature one or the other due to their independent correlation with others. If you notice, temperature has a low correlation with Humidity of 0.15, while dew point temperature has a mild correlation of 0.52. The same thing was spotted with solar radiation: dew point has a lower correlation of 0.10 and temperature a greater correlation of 0.36. The different correlations each variable possesses with another could be an advantage for certain models or procedures, so we are going to leave it as is. Every other feature does not possess an interesting correlation.

the daily curve, and the low coefficient for Hour is particularly revealing: a variable with two peaks and a valley between them has almost no monotonic trend for a Pearson coefficient to detect, even though it is the single most visible driver of demand in the entire dataset. These coefficients therefore measure linear association only, and their weakness is itself an indication of the shape of the relationships, setting an expectation to contrast against the feature importances reported later by the tree models.

The final verdict is that the dataset will be left as is, since imperfection will always be expected, especially when forecasting this kind of problem; the specific selection of data will create a biased output which we will not be able to take advantage of, as we have managed to perfect our problem. What is important to document is that all of this imperfection, although acceptable to proceed, must be explained rather than avoided so it can provide possible explanations further in this project. Evaluating my strategy for split, the seasonality-based division did succeed in giving most of the subsets representation for all seasons. It is important to mention that it failed during the autumn season since the training subset won't get any of the most extreme values, therefore creating the actual risk of training a model that has a bias due to omission. So no further study will be made; no variables will be removed, created or transformed. We are ready to enter the discussion of the model design.

#### IV. ML MODELS

##### A. The choosing of the model

The EDA established two facts that give us the clue for a preferred model choice. One is the relationship between several features and the non-linear demand. It contains many behaviours that seem cyclic or approach fuzzy classes, and the dependent variable itself does not seem to hold a trend to be described linearly. The other fact is that the ETL output we provided is not friendly with a linear model: take the hours as integers from 0-23; that is not thought of as cyclic, and 23 won't be mathematically thought of as near 0; same thing for seasons. The expected output of a linear regression will be a not-so-well-fitted line, as all of these factors prove to be of theoretical complexity for the regression model. We will produce, however, a linear regression model by hand. The idea will be that our starting point will be the final metrics and scores for the linear regression trained with batch gradient descent without a learning framework for academic purposes to learn more about the construction of the model. This first model is chosen because it is the easiest hypothesis to combine all 12 features we are creating; it will be the most natural first attempt. The OLS implementation is chosen over a plain framework regression because it reports the standard error, t statistic, p value and confidence interval of every coefficient. This allows judging whether each variable contributes meaningfully to the model rather than only measuring the fit as a whole, and provides a formal diagnosis of the multicollinearity anticipated by the correlation matrix. Secondly, we can run an OLS regression model to back up the performance of the

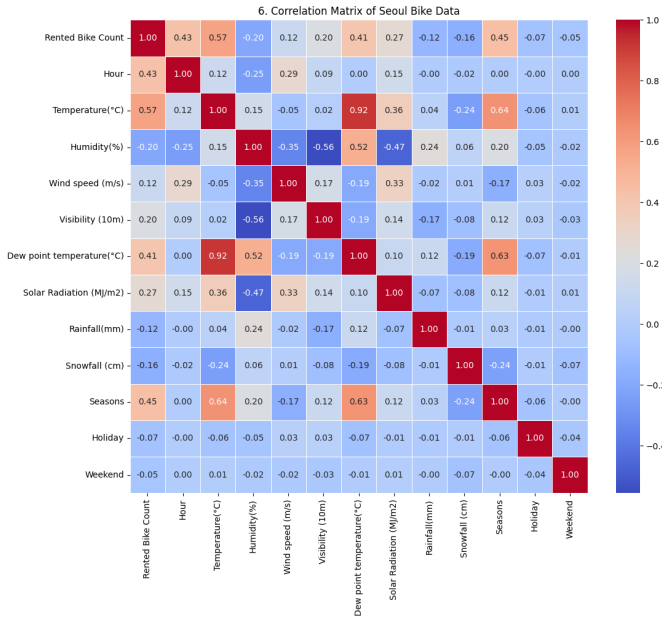


Fig. 10. Correlation matrix of the training subset.

Regarding the dependent variable, the strongest linear associations are Temperature at 0.57, Seasons at 0.45 and Hour at 0.43, while rainfall, snowfall, holiday and weekend show almost none. This does not mean the latter are uninformative. The line plots showed holiday and weekend clearly reshaping

model on the expected outputs and evaluate how much the linear model could really explain. This will also provide the expected records to beat; I have indicated that other models might fit this problem best, and then it shall be compared to.

Then our next models will compare something different to the linear thought. The proposals are a decision tree that performs regression instead of classification; this will be fitted first because it makes the change in assumption easy to follow instead of fitting one constant fixed weight per feature. In theory, I think this model will be able to pick up and isolate specific peaks and patterns that a linear coefficient could not represent. Then, by the same logic, I will try once again through a random forest regression with many more trees to reduce the variance that causes a single deep tree to overfit, by averaging many trees whose individual errors are uncorrelated. We are left then with four final models: the linear model by hand that will provide the floor to compare to, the OLS regression that will back up the performance of the linear and then the decision tree and random forest that will try to prove the question left open by the exploratory analysis. Since every model receives the same training subset, carrying the same documented gaps in autumn snowfall and summer rainfall, any difference in their results cannot be attributed to the split. The comparison therefore measures the cost of the linear assumption specifically. Finally it is important to mention that the four models are evaluated with the same metrics: RMSE and MAE, both expressed in bicycles, and  $R^2$ , the proportion of variance in hourly demand that the model explains. They are additionally compared against a trivial baseline that always predicts the mean demand of the training subset, since RMSE and  $R^2$  are not interpretable in isolation without a reference for what constitutes a useful prediction.

## V. LINEAR REGRESSION RESULTS

### A. Configuration and hyperparameter selection

We trained the model for 300 epochs with a learning rate of 0.05, using batch gradient descent over the 7,896 training instances. All thirteen parameters—the bias and one weight per feature—were initialised randomly in the range  $[-0.01, 0.01]$  under a fixed seed so that the run is completely reproducible. Random initialisation isn't actually necessary for this model: because the cost function of a linear regression is convex and has a single global minimum, the starting point affects the path we take but not the final destination. We mostly just adopted it as a general practice carried over from models where the choice really does matter.

We picked the number of epochs by eye rather than by applying a strict rule. Graph 7.1 shows that the cost begins to flatten out after roughly the first 50 epochs and that the change becomes barely perceptible after epoch 250, which seemed to confirm that 300 would be plenty. A more disciplined alternative, and one worth implementing in future iterations, would be to use early stopping: just monitoring the last ten epochs and stopping the training when the error no longer decreases by more than a set threshold, rather than fixing the budget in advance.

The learning rate was reached through trial and error. An earlier version of this implementation used larger values, which resulted in a chaotic loss curve that descended at first and then spiked around epoch 80. This is the classic signature of divergence, where each step overshoots the minimum and lands further away than it started, so we fixed it by reducing the learning rate. Going with even smaller values combined with a larger epoch budget would likely refine the result further, though obviously at the cost of longer training times.

### B. Learning curve

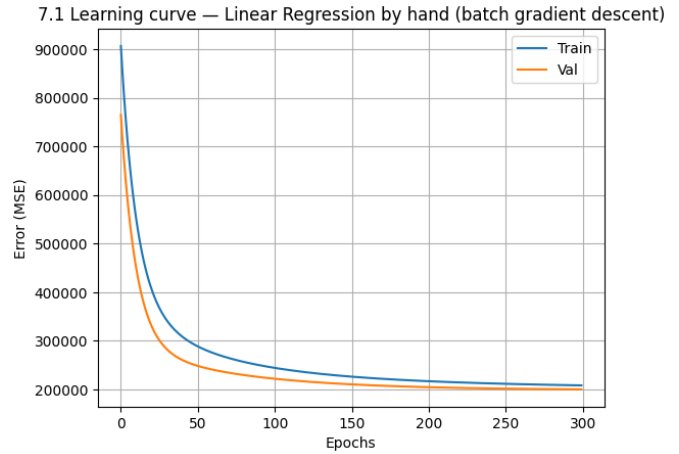


Fig. 11. Learning curve of the linear regression trained by hand.

Graph 7.1 shows the cost on both the training and validation subsets at every epoch. Both curves descend together and remain close throughout, with no widening gap between them. This is neither evidence of overfitting nor underfitting, a model that had begun memorising the training subset would show the training cost continuing to fall while the validation cost stagnated or diverged upward, and the absence of that pattern is expected for a hypothesis with too little capacity to memorise 7,896 instances through twelve weights. No regularisation strategy was therefore required for this model.

One detail deserves attention: the validation curve stays consistently below the training curve. Since the model never used validation data to update its weights, this does not indicate leakage. It indicates that the validation subset happened to be easier to predict than the training subset, which is plausible given that training carries the most extreme weather instances documented in the exploratory analysis, while validation covers only four days per season.

### C. Fitted parameters

Table VIII shows the final weights. The features with the biggest impact correspond to Hour (592.52), Temperature (436.94) and Seasons (339.20), with Humidity right behind in fourth place at -338.63. You can't read these numbers directly as bikes per unit, because the features were mean-normalised before training, but you can compare them against each other because every feature was brought to a similar range.

TABLE VIII  
PARAMETER RESULTS INCLUDING BIAS.

| Parameter                 | Weight    |
|---------------------------|-----------|
| Bias                      | 739.8380  |
| Hour                      | 592.5202  |
| Temperature(°C)           | 436.9375  |
| Humidity(%)               | -338.6270 |
| Wind speed (m/s)          | 88.6116   |
| Visibility (10m)          | 246.0339  |
| Dew point temperature(°C) | 282.0122  |
| Solar Radiation (MJ/m2)   | 157.3056  |
| Rainfall(mm)              | -35.7085  |
| Snowfall (cm)             | -31.4478  |
| Seasons                   | 339.2028  |
| Holiday                   | -67.1860  |
| Weekend                   | -67.8890  |

All the signs make logical sense. Hour and Temperature drive up demand, which lines up nicely with rentals growing through the day and warmer conditions encouraging cycling. Humidity reduces it, just as we'd expect. Visibility has a positive effect, meaning clearer conditions favour rentals. Dew point follows the same direction as Temperature; this makes perfect sense given that the two are physically linked and we already saw they correlate at 0.92. Solar Radiation is positive, matching the growing tendency we noticed during the first part of the day. Rainfall and Snowfall are negative, along with Holiday and Weekend, which all fits the behaviour we saw in the line plots.

The strongest confirmation comes when we compare these signs against the first row of the correlation matrix in graph 6.1. Every single one of the twelve weights lines up in direction with its corresponding correlation coefficient: Hour at 0.43, Temperature at 0.57, Humidity at -0.28, Seasons at 0.45, Rainfall at -0.12, Snowfall at -0.16, and so on. The magnitudes differ, of course, since the weights are the result of a joint optimisation while the correlations are just pairwise, but the fact that they completely agree in direction shows that the optimisation picked up on the real relationships we found during the exploratory analysis, rather than just pulling arbitrary values.

#### D. Metrics

TABLE IX  
LINEAR REGRESSION RESULTS.

| Subset     | MSE         | RMSE     | MAE      | R2     |
|------------|-------------|----------|----------|--------|
| Validation | 200087.3691 | 447.3113 | 330.7427 | 0.3996 |
| Test       | 255332.2068 | 505.3041 | 361.2471 | 0.3523 |

Table IX reports the final metrics. On the test subset the model reaches an  $R^2$  of 0.3523, an RMSE of 505.30 bikes and an MAE of 361.25 bikes. When compared to the baseline, which always predicts the training mean and reaches an RMSE of 628.21 with an  $R^2$  of -0.001, the model does seem to extract

real signal: it explains roughly 35% of the variance in hourly demand rather than none. Still, it is far from being a highly accurate predictor. An average error of 505 bikes is quite large when you consider that the training mean is 701 bikes per hour. Basically, the typical prediction is off by an amount similar to the typical demand itself. Then validation outperforms test on all four metrics, with an  $R^2$  of 0.3996 against 0.3523 and an RMSE of 447.31 against 505.30. Since the model saw neither subset during training, this is not evidence of overfitting; it indicates that the two holdout periods differ in difficulty, with test apparently containing the harder instances. With only 384 and 480 instances respectively, a difference of this size also falls within the variation expected from such small samples. Looking at the relationship between RMSE and MAE is also quite interesting. RMSE is always greater than or equal to MAE because squaring the errors before averaging heavily penalises the largest ones; the ratio between them therefore tells us how much the error is driven by a few extreme cases. On validation, this ratio is 1.35, and on the test set it goes up to 1.40. This means the model is making a small number of very large mistakes, and it makes them more often on the test subset.

#### E. Prediction and limitations

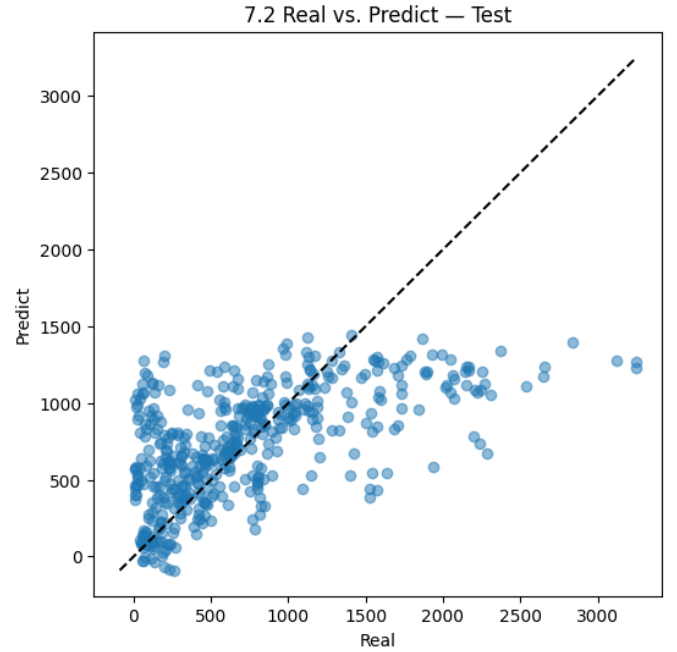


Fig. 12. Real vs predicted values, linear regression by hand (test subset).

Graph 7.2 makes the failure clearly visible. The predictions concentrate beneath 1,500 bikes, while the real demand reaches beyond 3,000. The reason is probably due to the underestimation of peak labour hours; high demand in volume like this is not only produced by outliers, which was a misconception of the model. The predictions below zero in the graph have no physical meaning for a count of rented bikes. This

was previously expected from our findings in the exploratory analysis: the model is unable to represent the labour peaks, as it is constrained to assign one constant weight to an hour variable encoded from 0 to 23. This limitation was the result of a poor decision regarding data transformation. The model settles on an intermediate value that minimises the squared error on average while remaining wrong at both extremes of the day. The results therefore confirm the prediction made in the previous subsection.

Finally, an observation that concerns the optimisation rather than the model itself is that the fitting of the same hypothesis through ordinary least squares (OLS), reported further into the report, yields a better fit on the exact same data. Since both procedures minimise the identical cost function, and that function is convex with a single minimum, the difference indicates that 300 epochs were not enough for the gradient descent to reach the theoretical optimum value. The assumption regarding the flattening of the loss curve was incorrect; although the curve, in my perception, appears flat at the scale of the plot, it tends to improve further into the epochs. The effect could be due to the concentration in the weights of the most skewed features, where dividing by the maximum reduces almost every value towards zero and produces small gradients to improve on those weights. The conclusions of this section are drawn against the OLS implementation, which is discussed in the next section.

## VI. OLS MODEL ANALYSIS

This section evaluates some of the decisions taken in the gradient descent implementation and establishes the theoretical limit of a linear approach to this problem. For all intents and purposes we will not focus in depth on the individual output of each coefficient, but on the overall model.

The final scores were an MSE: 234724.37761597117, RMSE: 484.4836195538206, MAE: 369.47152553336394, R2: 0.4046179522745339.

As mentioned before the comparison between OLS and the by hand made code make a clear difference on  $R^2$ , the OLS model scoring 0.4046 against 0.3523, higher with a better adjustment. This is plausible due to the insufficient number of epochs or even a general combination of the hyper parameters. The reason to believe this is once again, both models minimise the same cost function, they get the same dataset and globally in the function there are not minimal locals. Therefore it is attributable to the optimisation and not to the hypothesis, the ordinary least squares solves the problem in similar results with a closed form and arrives directly at the optimum that gradient descent is still approaching.

In the OLS Regression Results of Table A the model is tagged with a R-squared of 0.548 which doesn't serve to compare against the test score since it is the equivalent of the training phase, while the reported score is made on the test subset. What is important to notice is that the gap between the two is mild, which indicates that the linear model is not over-adjusting. In fact, it does not appear to have the capacity

to reach a high training score in the first place, which points to a structural limitation rather than a generalization problem.

As an overview, the parameters are not equal to the ones reported by our by-hand linear model. This can again be due to the state of convergence of the training, so what we can focus on instead is the validation of the coefficients. The odd one out is Dew point temperature. Its magnitude does not compare to the others, and although its standard error is low, a p value of 0.201 means there is not enough evidence to reject the possibility that the true coefficient is zero. In more detail, we discard its significance because its confidence interval crosses zero, ranging from -3.04 to 14.50. The connection with the heatmap indicates multicollinearity: we found earlier that Temperature and Dew point correlate at 0.92. Using both variables creates the risk that two predictors explain the same thing, which makes the coefficients come out in erratic ways and makes it difficult for the model to tell them apart and attribute the change to one variable or the other.

Checking the remaining signs, the expectation held for most features: Humidity, Rainfall, Holiday and Weekend all come out negative as the exploratory analysis anticipated. Three do not match. Visibility comes out at -45.05, suggesting that worse visibility increases demand; Solar Radiation at -85.63, suggesting that less sunlight increases demand; and Snowfall at +28.89, suggesting that more snow increases it. All three contradict both physical intuition and the direction reported by the by-hand model, which returned them as positive, positive and negative respectively. Sign reversals of this kind are a known consequence of multicollinearity, and they should not be read as physical findings but as additional evidence that the linear model is struggling to separate the contribution of each variable.

The overall conclusion is that roughly 40% of the variance in hourly demand is the ceiling of a linear model on this specific dataset. Reaching it required a closed form solution rather than gradient descent, but even at the optimum the model remains unable to represent the cyclical behaviour identified in the exploratory analysis. This establishes the record that the tree-based models are expected to beat.

TABLE X  
OLS COEFFICIENTS WITH STATISTICAL SIGNIFICANCE.

|                           | Coefficient | Std. Error | t        | $P >  t $ | CI 2.5%   | CI 97.5% |
|---------------------------|-------------|------------|----------|-----------|-----------|----------|
| const                     | 491.5820    | 108.2777   | 4.5400   | 0.0000    | 279.3291  | 703.8350 |
| Hour                      | 29.0794     | 0.7715     | 37.6910  | 0.0000    | 27.5670   | 30.5918  |
| Temperature(°C)           | 20.3562     | 4.2290     | 4.8134   | 0.0000    | 12.0662   | 28.6463  |
| Humidity(%)               | -10.4886    | 1.2023     | -8.7237  | 0.0000    | -12.8454  | -8.1317  |
| Wind speed (m/s)          | 20.8444     | 5.3595     | 3.8892   | 0.0001    | 10.3384   | 31.3504  |
| Visibility (10m)          | -45.0531    | 21.1270    | -2.1325  | 0.0330    | -86.4676  | -3.6386  |
| Dew point temperature(°C) | 5.7263      | 4.4738     | 1.2800   | 0.2006    | -3.0435   | 14.4962  |
| Solar Radiation (MJ/m2)   | -85.6291    | 8.0648     | -10.6176 | 0.0000    | -101.4382 | -69.8199 |
| Rainfall(mm)              | -59.2789    | 4.8928     | -12.1156 | 0.0000    | -68.8701  | -49.6878 |
| Snowfall (cm)             | 28.8874     | 14.5496    | 1.9854   | 0.0471    | 0.3663    | 57.4084  |
| Seasons                   | 128.2638    | 6.0664     | 21.1432  | 0.0000    | 116.3720  | 140.1556 |
| Holiday                   | -124.6892   | 21.6043    | -5.7715  | 0.0000    | -167.0392 | -82.3391 |
| Weekend                   | -82.5923    | 10.8583    | -7.6063  | 0.0000    | -103.8775 | -61.3071 |

OLS Regression Results Table A

|                           |                  |                     |           |       |          |         |
|---------------------------|------------------|---------------------|-----------|-------|----------|---------|
| Dep. Variable:            | y                | R-squared:          | 0.548     |       |          |         |
| Model:                    | OLS              | Adj. R-squared:     | 0.547     |       |          |         |
| Method:                   | Least Squares    | F-statistic:        | 795.9     |       |          |         |
| Date:                     | Sun, 13 Sep 2026 | Prob (F-statistic): | 0.00      |       |          |         |
| Time:                     | 23:38:13         | Log-Likelihood:     | -59164.   |       |          |         |
| No. Observations:         | 7896             | AIC:                | 1.184e+05 |       |          |         |
| Df Residuals:             | 7883             | BIC:                | 1.184e+05 |       |          |         |
| Df Model:                 | 12               |                     |           |       |          |         |
| Covariance Type:          | nonrobust        |                     |           |       |          |         |
| =====                     |                  |                     |           |       |          |         |
|                           | coef             | std err             | t         | P> t  | [0.025   | 0.975]  |
| -----                     |                  |                     |           |       |          |         |
| const                     | 491.5820         | 108.278             | 4.540     | 0.000 | 279.329  | 703.835 |
| Hour                      | 29.0794          | 0.772               | 37.691    | 0.000 | 27.567   | 30.592  |
| Temperature(°C)           | 20.3562          | 4.229               | 4.813     | 0.000 | 12.066   | 28.646  |
| Humidity(%)               | -10.4886         | 1.202               | -8.724    | 0.000 | -12.845  | -8.132  |
| Wind speed (m/s)          | 20.8444          | 5.359               | 3.889     | 0.000 | 10.338   | 31.350  |
| Visibility (10m)          | -45.0531         | 21.127              | -2.132    | 0.033 | -86.468  | -3.639  |
| Dew point temperature(°C) | 5.7263           | 4.474               | 1.280     | 0.201 | -3.044   | 14.496  |
| Solar Radiation (MJ/m2)   | -85.6291         | 8.065               | -10.618   | 0.000 | -101.438 | -69.820 |
| Rainfall(mm)              | -59.2789         | 4.893               | -12.116   | 0.000 | -68.870  | -49.688 |
| Snowfall (cm)             | 28.8874          | 14.550              | 1.985     | 0.047 | 0.366    | 57.408  |
| Seasons                   | 128.2638         | 6.066               | 21.143    | 0.000 | 116.372  | 140.156 |
| Holiday                   | -124.6892        | 21.604              | -5.772    | 0.000 | -167.039 | -82.339 |
| Weekend                   | -82.5923         | 10.858              | -7.606    | 0.000 | -103.878 | -61.307 |
| =====                     |                  |                     |           |       |          |         |
| Omnibus:                  | 1145.447         | Durbin-Watson:      | 0.523     |       |          |         |
| Prob(Omnibus):            | 0.000            | Jarque-Bera (JB):   | 2106.550  |       |          |         |
| Skew:                     | 0.928            | Prob(JB):           | 0.00      |       |          |         |
| Kurtosis:                 | 4.720            | Cond. No.           | 1.43e+03  |       |          |         |

Notes: [1] Standard Errors assume that the covariance matrix of the errors is correctly specified. [2] The condition number is large, 1.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

## VII. DECISION TREE (FRAMEWORK)

For the decision tree we chose a maximum depth of 10. The depth is the number of chained questions a branch can ask before reaching a conclusion, so the objective is to find a point deep enough for the tree to capture the structure of the data, but not so deep that it becomes specific to individual training instances and simply memorises them. The depth of 10 was selected by fitting a tree at each of a set of candidate depths and comparing their performance. As graph 8.1 shows, the training score approaches a perfect fit as depth increases, which is actually counterproductive rather than desirable, this would mean the tree is memorising the training instances. The validation score reveals this, since it peaks at depth 10 and declines beyond that point, while below it neither subset reaches an acceptable score. Depth 10 is therefore the point at which the tree understands the data without becoming too specific to it, among the depths tested. One consideration worth mentioning is that the candidate set jumped from 10 to 12, so an intermediate value of 11 remains untested and could be worth exploring. The results on the test subset were an MSE of 67,713.96, an RMSE of 260.22 bikes, an MAE of 166.29 bikes and an  $R^2$  of 0.8282. Against the 0.4046 obtained by the OLS regression on the same data, the tree more than doubles the proportion of variance explained, and reduces the average error from 484 to 260 bikes.

To understand these outputs better we turn to graph 8.2, which shows the top levels of the final tree. The root splits on Temperature, which was actually a bit unexpected: the hour of the day seemed like the most likely candidate given what we saw in the exploratory analysis. It makes total sense, though, since splitting on temperature first separates the dataset into two regimes that behave differently, and the tree can then explore seasonality on one side and the hourly cycle on the other. In the top levels, the variables selected are Temperature, Seasons, Hour, Solar Radiation, Humidity and Rainfall, with the remaining features used further down to refine the details for more specific scenarios and tune better predictions. The most recurrent variable in the top levels is Hour, which lines up perfectly with what we found in the exploratory analysis, where we identified it as a key driver of demand. The interesting part isn't just that it appears, but that it keeps showing up along a single path: the tree asks about the hour several times in a row, narrowing down the window at each step until it isolates a specific range, like the evening commute. This is exactly what a linear model can't do. A single coefficient for Hour has to describe the entire day with one constant rate of change, whereas the two peaks we saw in graph 4 require the model to treat different parts of the day differently. The tree pulls this off through nested thresholds rather than relying on a rigid functional form.

We have only graphed the first few levels, since the full tree at depth 10 has way too many nodes and is unreadable. It's also worth pointing out that this tree isn't built using entropy. Entropy and Gini are impurity measures for classification problems. In the nodes shown in graph 8.2, the reported

quantity is the squared error, and at every split, the tree looks for the threshold that most reduces the variance of the target inside the resulting child nodes. Also important to mention that unlike gradient descent, a decision tree is not trained iteratively, instead it's built in a single recursive pass, choosing the best split at every node. That is why there is no loss curve to watch over epochs in these models. The equivalent diagnostic is the graph 8.1 which plots  $RSquared$  against maximum depth for the training and validation subsets and serves also the purpose of revealing where the model stops generalising.

## 8.2 Decision Tree — first levels

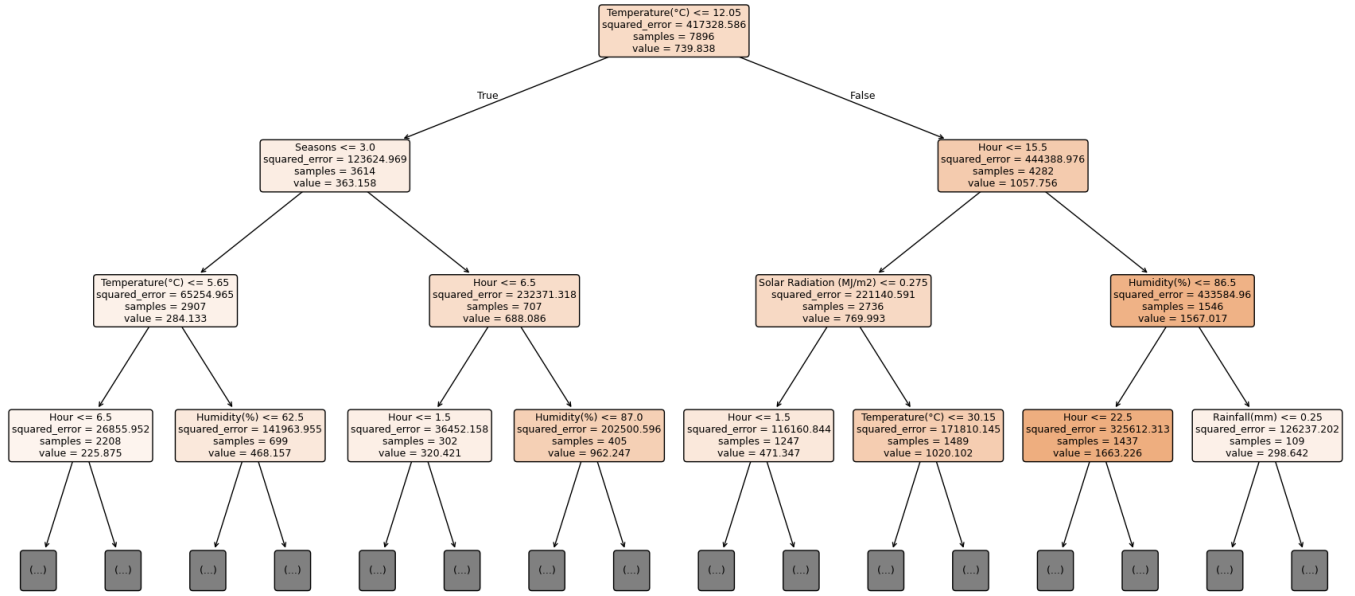


Fig. 13. First levels of the decision tree.

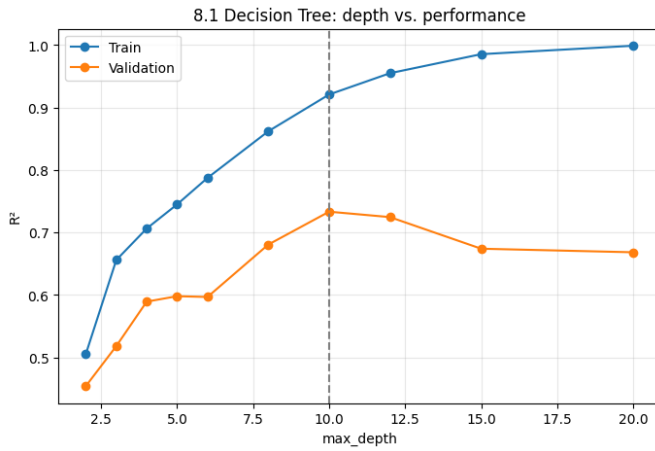


Fig. 14. Decision tree performance against maximum depth.

## VIII. RANDOM FOREST MODEL

The reason for building a random forest rather than settling for a sufficiently accurate decision tree is to extend the same concept. A single tree is prone to overfitting and memorisation, a behaviour observed in graph 8.1 at depths of 12 and beyond. The random forest addresses this through two sources of randomness that make its 200 trees differ from one another. The first one is the use of bootstrapping. Instead of training every tree on the same instances, each one will receive a random sample drawn with replacement from that set so every tree sees some repeated rows and others not at all. The second is the feature subsampling, at each split the tree considers only a random subset of the twelve variables rather than all of them. The combined effect is that the trees memorise different things. Their individual errors are supposed to be uncorrelated and when the forest produces a prediction it averages the output of all 200 trees. It is expected those errors will cancel out while the signal they share survives. This is also why the forest doesn't restrict depth the way the single tree does. Each tree is grown fully and is deliberately allowed to overfit; the control comes from the averaging rather than from constraining any individual tree. On the test subset, the forest scored an MSE of 41,250.11, an RMSE of 203.10 bikes, an MAE of 133.24 bikes and an  $R^2$  of 0.8954. This is a nice step up from the single tree, which hit 0.8282, and it more than doubles the 0.4046 we saw from the OLS regression. Looking at it in absolute terms, the average error drops from 484 bikes with the linear model down to 203 with the forest. On the training subset, the forest hits an  $R^2$  of 0.9890 compared to 0.8954 on the test set. This gap is actually totally expected rather than alarming, and it follows directly from the mechanism we just described: because every individual tree was allowed to fit the training data completely, a high training score is essentially there by design. What really matters is that the test performance holds strong at 0.8954, which confirms that the averaging is actually doing its job. A forest that was genuinely failing to generalise would show a similar training score alongside a complete collapse on the test set

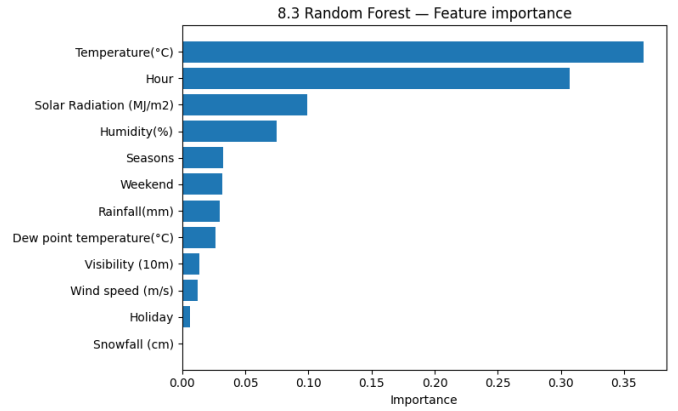


Fig. 15. Random forest feature importance.

Graph 8.3 shows the importance of each feature. Temperature, Hour and Solar Radiation drive most of the model's decisions, and the ranking differs from what the correlation matrix would have predicted. Temperature leads in both, but Solar Radiation rises from sixth place by correlation at 0.27 to third by importance, while Seasons falls from second at 0.45 to fifth. It is worth paying special attention to Hour, whose correlation of 0.43 is only moderate yet ranks second in the forest. This is not a contradiction: a correlation coefficient only measures linear association, and a variable with two peaks separated by a valley has very little monotonic trend for it to pick up on. The forest is not restricted to linear patterns, so it can exploit predictive value that the coefficient largely missed. The drop of Seasons is explained by the opposite effect: its information overlaps heavily with Temperature, so once the forest splits on temperature it has little left to gain from the season label. This same logic explains why the earlier linear models failed, since they could only access the portion of each relationship that a coefficient could see.

When we compare the importances from the single tree against those of the forest, the ranking is essentially the same. Since the two models were fitted using entirely different procedures, this agreement suggests that the ranking reflects the true underlying structure of the data rather than just a quirk of how one specific model was fitted.

## IX. CONCLUSION

### A. Very brief summary.

One year of hourly bike-rental records from Seoul was processed through an ETL that encoded categorical variables, scaled visibility to a ratio, and imputed the 295 hours in which the system was closed. The data was then divided into training, validation, and test subsets drawn systematically from the last days of every season.

An exploratory data analysis characterised the demand, identifying two daily peaks driven by the working schedule, and documented the extreme weather instances that the split inadvertently left out of the training set.

Four models were then fitted on that identical training subset: a linear regression implemented from scratch with batch gradient descent, an Ordinary Least Squares (OLS) regression that established the analytical ceiling of a linear hypothesis and reported the statistical significance of each coefficient, a decision tree whose depth was selected on validation, and a random forest of 200 trees.

All were evaluated using RMSE, MAE, and  $R^2$  against a naive baseline that always predicts the training mean. Ultimately, the linear models explained roughly 40% of the variance in hourly demand, while the random forest captured 89.5%—a stark contrast that isolates the linear assumption itself, rather than the data split, as the true limiting factor.

### B. Versions and Comparison

TABLE XI  
METRICS FOR EVERY MODEL ON THE TEST SUBSET.

| Model                   | RMSE    | MAE     | R2     |
|-------------------------|---------|---------|--------|
| Baseline (mean)         | 628.215 | 490.529 | -0.001 |
| Linear Regression (OLS) | 484.484 | 369.472 | 0.405  |
| Decision Tree           | 260.219 | 166.287 | 0.828  |
| Random Forest           | 203.101 | 133.243 | 0.895  |

Table XI orders the four models by their performance on test subset scores. The trivial baseline calculated by predicting the training mean reaches an  $R^2$  of -0.001 which is supposed to be the foundation to scale better the scores of the three models and that the simple average won't predict anything. The linear regression trained by hand reaches 0.352 but considering the OLS implementation we should be able to reach a top of 0.405 since it solves the problem in closed form. The decision tree reaches 0.828 and random forest performs even better getting 0.895. Is important to mention once again is that the gap between linear models is 0.05, attributable to the convergence of gradient descent not to a different approach or wrangle of data. The difference between the best linear model and the single decision tree is 0.42, which is a substantial increase in performance and the random forest differs from the tree by 0.07. The progression on scores is not gradual there is a noticeable change between the two families of models which the gap is shown as almost eight times more than any refinement that could be made within the linear family. Considering the errors the baseline is generally wrong by 628 bikes in an instance and the linear model is wrong by 484 which is a very low performance in the real application. A mistake still a poor result in practical terms, though a clear improvement over the baseline and finally the decision tree wrong about 260 bikes and random forest about 203 bikes in average per hour. Against the training mean of 701 bikes per hour, the linear models typical error is considerable to the demand itself, the forest reduces it less to a third of that. The improvement is very desirable for an operator deciding how many bicycles to redistribute to a station, that is the difference between figure that cannot be acted upon and one that can. It

is important to mention, that even so, an average error of 203 bikes per hour remains considerable in operational terms. An  $R^2$  of 0.895 describes how much of the variance the model explains, not whether the residual error is small enough to act on, and for a station deciding its exact allocation those 203 bikes are still a wide margin. Within the scope this project covers it is the best result obtained, but it should not be read as a solved problem.

TABLE XII  
METRICS FOR EVERY MODEL ON VALIDATION AND TEST.

| Model                   | Subset | MSE         | RMSE     | MAE      | R2      |
|-------------------------|--------|-------------|----------|----------|---------|
| Baseline (mean)         | val    | 335166.7932 | 578.9359 | 464.5786 | -0.0058 |
|                         | test   | 394653.6426 | 628.2146 | 490.5287 | -0.0010 |
| Linear Regression (OLS) | val    | 196734.1977 | 443.5473 | 342.3645 | 0.4096  |
|                         | test   | 234724.3776 | 484.4836 | 369.4715 | 0.4046  |
| Decision Tree           | val    | 88885.2143  | 298.1362 | 179.7246 | 0.7333  |
|                         | test   | 67713.9596  | 260.2191 | 166.2874 | 0.8282  |
| Random Forest           | val    | 61805.4604  | 248.6070 | 153.8811 | 0.8145  |
|                         | test   | 41250.1054  | 203.1012 | 133.2425 | 0.8954  |

Table XIII compares the feature importances of both tree models. The rankings for feature are almost identical. The temperature leads on decision tree with 0.3629 and random forest with 0.3654, Hour follows at 0.3335 and 0.3073, and Solar Radiation is third at 0.0953 and 0.0991. Since both models were fitted by different procedures, one single tree limited by a depth of 10 and the other an ensemble of 200 unrestricted trees, lead us to the conclusion that the ranking reflects the true structure of the data rather than a particular opinion of fit of just one model.

The EDA closed with a question it could not answer, whether the seasonality based split had removed important information the model needed, since the training subset received no autumn snowfall and fewer summer rain instances than validation or test. The comparison answers it. Every model was trained on that identical subset, carrying those identical gaps. If the split had been the limiting factor, no model could have surpassed the ceiling it imposed. The random forest reaches 0.895 under the same constraints that left the linear models at 0.40, which isolates the linear assumption as the dominant limitation rather than the data division. The split remains a documented weakness of this work, but a secondary one.

This does not make the linear model a wasted effort. Implementing it from scratch made the mechanism of gradient descent explicit rather than assumed, and its failure was not a dead end but the evidence that motivated the tree-based models. The compression towards the centre visible in Graph 7.2, the sign reversals produced by multicollinearity in the OLS coefficients, and the ceiling of 40% were what turned the suspicion raised by the exploratory analysis into a measured quantity.

Graphs 8.4 and 8.5 make this difference visually obvious. The compression towards the centre that characterised the linear model has completely disappeared, the predictions now closely follow the diagonal across the entire range—including

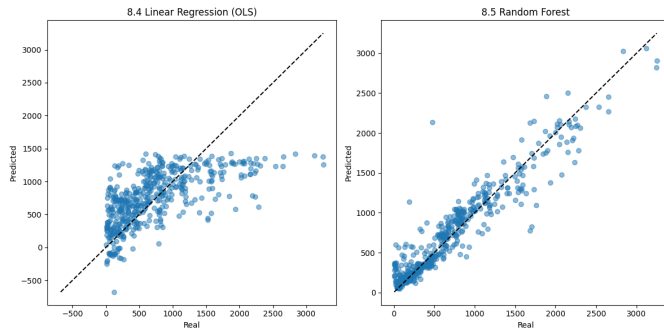


Fig. 16. Real vs predicted values, OLS against Random Forest (test subset).

those high-demand peak hours that the linear model systematically underestimated—and they finally no longer fall below zero.

### C. Critique of the decisions taken

TABLE XIII  
FEATURE RANKING BETWEEN THE DECISION TREE AND THE RANDOM FOREST.

| Feature                   | Decision Tree | Random Forest |
|---------------------------|---------------|---------------|
| Temperature(°C)           | 0.3629        | 0.3654        |
| Hour                      | 0.3335        | 0.3073        |
| Solar Radiation (MJ/m2)   | 0.0953        | 0.0991        |
| Humidity(%)               | 0.0881        | 0.0749        |
| Seasons                   | 0.0358        | 0.0323        |
| Weekend                   | 0.0335        | 0.0320        |
| Rainfall(mm)              | 0.0204        | 0.0301        |
| Dew point temperature(°C) | 0.0173        | 0.0262        |
| Visibility (10m)          | 0.0046        | 0.0134        |
| Wind speed (m/s)          | 0.0038        | 0.0125        |
| Holiday                   | 0.0040        | 0.0061        |
| Snowfall (cm)             | 0.0007        | 0.0007        |

The decision with the largest cost in the encoding was the temporal features. Representing hour as integer did the trick for differencing and telling apart different hours however the behaviour is cyclic, after 23 hours the next day starts with 0 and therefore they are adjacent. 0-23 tells a linear model that hour 23 is not even close to hour 0 and are the most distant points of the day. mpt consecutive. This forces a single constant rate of change across a curve that the exploratory analysis had already shown to have peaks. Same case to the seasons encoded from 1 to 4, it once again imposes an ordering and magnitude difference purely on categorical variable. This choice cost the linear model a substantial part of their performance and cost the tree models nothing, the threshold split is indifferent to this scale of the label it splits on. If this project had remained only with linear models this would have been a first thing to correct. A must do if you want to recreate it.

The choice of keeping both temperature and dew point despite their high correlation of 0.92 proved to be a mixed decision. For the linear model it was clearly harmful as it produced a visible non-significant coefficient in OLS regression and contributed to the sign reversals in other features. It is also

a must do if you want to recreate it or work only with linear models as you will ultimately tangle with multicollinearity problems. For the tree models it was in fact harmless since Dew point simply received a low importance and the forest relied on temperature instead. The reasoning given in the exploratory analysis, that each variable correlates differently with the remaining features and might therefore contribute something distinct, held for the trees but not for the regression.

The seasonality based split achieved its main objective, which was to give every subset a representative and equitable data characteristics across all four seasons rather than concentrating the holdout in a single period. However it failed in the specific cases documented in Section III, and the failure was reported rather than ignored. The limitation is real but bounded and the model comparison showed it was not what determined the results.

The choice of 300 epochs for the manual implementation was also a clearest error. It was based on visual inspection of the loss curve which appeared flat, and the OLS regression later showed that the parameters were still improving. A curve can look flat at the scale of a plot while the optimisation continues to progress, and no conclusion about convergence should rest on appearance alone. Early stopping monitored against a numerical threshold, rather than a fixed budget chosen by eye, would have avoided it. It is a must fix to try more epochs or implement an automatic stop epoch.

The imputation of the 295 non-functioning hours remains firmly defensible. Those original zeros reflected an absence of service rather than an actual absence of demand. Feeding them into the model unchanged would have taught it a false rule based on whatever random weather conditions happened to coincide with the outages—especially since the Functioning Day variable was excluded from the final feature set. Furthermore, because these specific instances remain clearly identifiable through the mapping subset, the decision is fully auditable and reversible rather than a permanent distortion of the raw data.

### D. Final considerations.

The first topic of further work concerns the data split itself. Drawing the holdout by individual hours rather than by whole days—while enforcing rules to guarantee representation across morning, daytime, and night periods—should produce far more comparable subsets. Features change much less between adjacent hours than across whole days, meaning the coldest or rainiest instances of the year would be far less likely to fall entirely outside the training subset, ensuring each split carries a more even share of the extremes.

This shift would particularly benefit the linear models, whose performance suffered the most from the omissions documented in our exploratory analysis. However, this proposal carries a crucial risk: with chronologically ordered data, assigning hour 14 to training and hour 15 to testing places two nearly identical instances on opposite sides of the division, which artificially inflates apparent performance. Any such arrangement would strictly require a minimum chronological

gap between subsets to keep the evaluation honest. An idea proposal would be to produce a random but equitable split in instances for the training, validation and test.

The second improvement concerns the representation of cyclical behaviour. The daily cycle appears in almost every feature examined—a pattern that sine and cosine functions describe naturally, whereas a raw integer encoding cannot. Transforming the Hour variable into its sine and cosine components would make hour 23 and hour 0 adjacent in the feature space, exactly as they are in reality. This would grant linear models access to a continuous structure they currently cannot represent at all.

It is worth noting exactly where this adds value: the linear family would benefit substantially, whereas tree models would see very little gain, as they already efficiently isolate hour ranges through nested thresholds. The broader takeaway is that applying targeted mathematical transformations before training can shoulder part of the modelling burden, serving as a much more principled remedy for the limitations observed here than standard normalisation alone.

Finally, the structural limitations of tree-based models deserve specific attention. Decision trees and random forests cannot extrapolate: outside the boundaries observed during training, they simply predict the value of the nearest leaf, which remains constant. If a winter colder than any in the training subset were to occur, the forest would merely return the prediction associated with the coldest range it has already seen, whereas a linear regression would at least extend its mathematical trend. While the forest is the stronger model on this dataset, its predictive advantage is strictly confined to the range its data covers.

This limitation means the model could not be expected to perform equally well on present-day data. Recent years have brought noticeable shifts in Seoul's weather behaviour, introducing unprecedented meteorological events. Because the forest is completely constrained by its historical training set, encountering these never-before-seen extremes would inevitably result in inaccurate, flattened predictions. This vulnerability reinforces the need for a genuinely representative training subset and makes a strong case for integrating multiple years of records to capture the inter-annual climate variations that a single cycle simply cannot show. methodological gap this report leaves open.

#### *E. Closing remarks.*

The objective stated at the beginning was to predict hourly bike rental demand in Seoul and to determine which family of models suits the structure of this problem. Both were met. The random forest explains 89.5% of the variance in hourly demand with an average error of 203 bikes, and the comparison established that the limitation of the linear models was their hypothesis rather than the data they received.

If this system were to be deployed, the random forest would be the model delivered, accompanied by an explicit caveat regarding its inability to extrapolate. However, the linear model would still be worth retaining alongside it—not

as a competitor, but as a crucial diagnostic tool. It offers the interpretable coefficients and formal significance tests that the forest cannot provide, which are exactly what allow a practitioner to explain why a prediction is what it is.

Ultimately, this project made clear that the decisive question was never which algorithm performs better in general, but whether the assumptions of a model match the underlying shape of the data. The exploratory analysis identified that shape long before any model was fitted, and every result that followed confirmed what it had anticipated. The most valuable outcome of this work was not the final performance metric, but the rigorous process of documenting each limitation as it appeared—ensuring that the results could be fully explained, rather than merely reported.

#### *F. Acknowledgements.*

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